

Dietary Intake Is Associated with Phthalate Body Burden in a Nationally Representative Sample

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Background Phthalates are compounds that are used in a wide range of consumer products. However, the contribution of dietary intake to phthalate exposure has not been well defined. **Objective** The objective of this study was to assess the contribution of different food types to phthalate exposure. Phthalates are chemicals of concern because of the high levels measured in people and the environment, as well as the demonstrated toxicity in animal studies and limited epidemiological studies. Previous research, although limited, has suggested that phthalates contaminate food in various countries. **Methods** We conducted an exploratory analysis of data collected as part of the 2003–2004 National Health and Nutrition Examination Survey (NHANES). Associations between dietary intake (assessed by a 24-hr dietary recall) for a range of food types (meat, poultry, fish, fruit, vegetable, and dairy) and phthalate metabolites measured in urine were analyzed using multiple linear regression modeling. **Results** We found that metabolites of di-(2-ethylhexyl) phthalate (DEHP) and high-molecular-weight phthalate metabolites were associated with the consumption of poultry. Monoethyl phthalate, the metabolite of diethyl phthalate (DEP), was associated with vegetable consumption, specifically tomato and potato consumption. **Discussion** These results, combined with results from previous studies, suggest that diet is an important route of intake for phthalates. Further research is needed to determine the sources of food contamination with these toxic chemicals and to describe the levels of contamination of U.S. food in a large, representative U.S. sample.